

3) Optional: Relaxation Oscillator

Purpose of the laboratory work:

- Analyze and describe the operation of the relaxation oscillator described below (refer to handouts and/or electronics book)

The relaxation oscillator uses an op-amp with positive feedback to generate periodic waveforms. The circuit is a bistable multivibrator, where the output switches between two stable states based on capacitor charging. As the capacitor charges through a resistor, the voltage reaches a threshold, causing the op-amp to switch states and produce a square wave. The period of oscillation depends on the resistor and capacitor values.

- Calculate the values of the components so that the circuit has an oscillation frequency of 100 [Hz], and include the values of R_1 , R_2 , R and C in the report

As in the first experiment $R_1 = R_2 = 1 [k\Omega]$ to achieve symmetry and $\beta = \frac{R_1}{R_1 + R_2} = 0.5$.

The characteristic time is given by the period and therefore we can apply the formula:

$$T = \frac{1}{f} = 2\tau \ln \frac{1 + \beta}{1 - \beta} \rightarrow \tau = RC = 4.55 \cdot 10^{-3} [s]$$

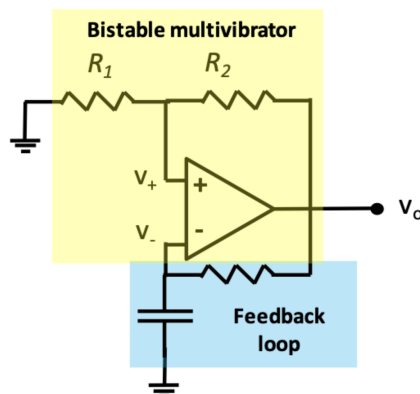
An real capacity is 1 [μF] and therefore the resistance is $R = 4.5 [k\Omega]$

- Add the waveforms of v_- , v_o , v_+ to the report



Oscilloscope

- Save the data related (txt or csv) to the waveform of v_- , and derive the time constants of capacitor charge and discharge. Check whether the values match the theoretical ones



Circuit Diagram



Oscilloscope Charge/Discharge

From the oscilloscope it is possible to directly measure the time difference between the maximum and minimum voltages at the capacitor. Due to time limitations only the picture is available and the distance is ≈ 2.5 squares. This equals to ≈ 5 [ms] which corresponds with our theoretical result of 4.77 [ms].